Number Systems
and
Base Conversions

- **Data** can be defined as a representation of facts, concepts, or instructions in a formalized manner, which should be suitable for communication, interpretation, or processing by human or electronic machines.

- **Data** is represented with the help of characters such as alphabets (A-Z, a-z), digits (0-9), or special characters (+,-,/,*,<,>,=, etc.)

- **Information** is organized or classified data, which has **some meaningful values for the receiver**. Information is the processed data on which decisions and actions are based.

- Collection, manipulation, and processing of collected data for the required use is known as **data processing**. It is a technique normally performed by a computer;



Suppose think about **a robot**, which can sense surrounding sounds and voices, and after collecting surrounding sounds, It will find out the "Play" sound from those sounds and will do a specific task. So then,

Data: Data are the signals, sounds, voices, or noises around the robots. It receives those sounds from its surroundings through sensors.

Information: When it processes those data and recognizes the specific sound 'Play', then this sound "Play" will become an information for that robot.

Knowledge: Now it will do as the instructions are programmed inside it for the information "Play". So, the program is the knowledge for the robot.

Wisdom: The knowledge we have, humans, we have given instructions to that robot to perform a specific task with that sound. We are controlling that robot, So, we are in the wisdom level.

Common Number Systems

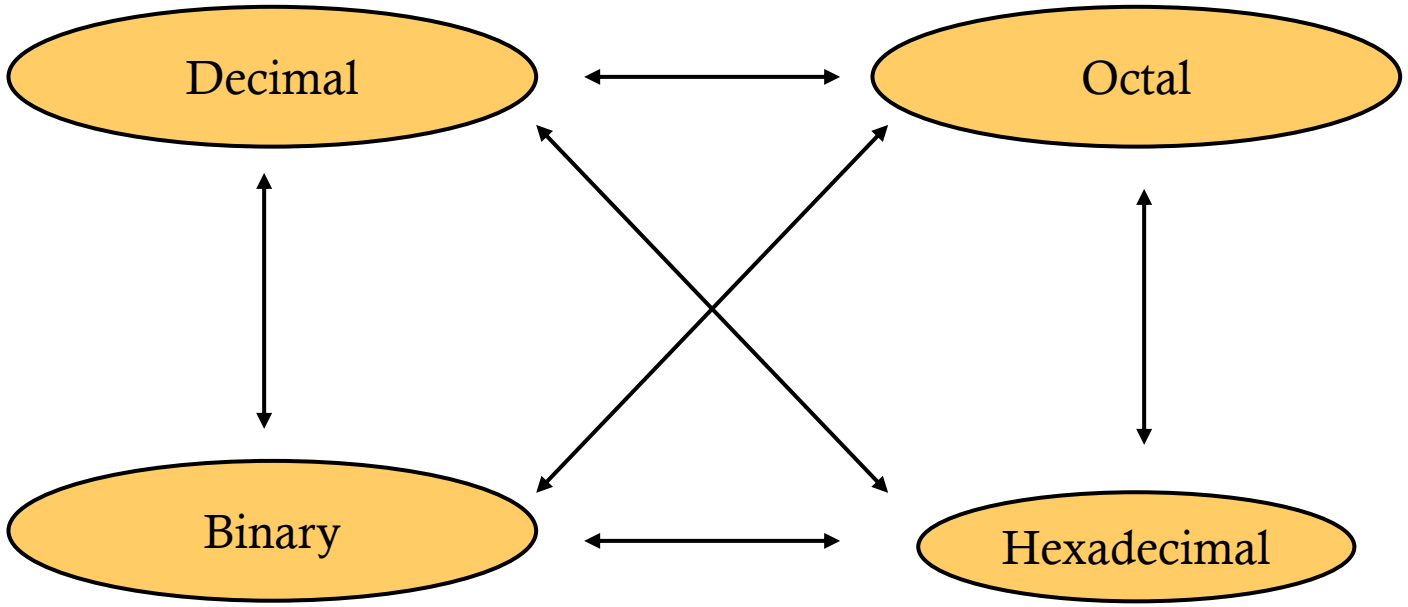
System	Base or Radix	Symbols	No. of Digits	Used by humans?	Used in computers?
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	10	**Yes**	No
Binary	2	0, 1	2	No	**Yes**
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	8	No	No
Hexa-decimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	16	No	No

Decimal	Binary	Octal	Hexa-decimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7

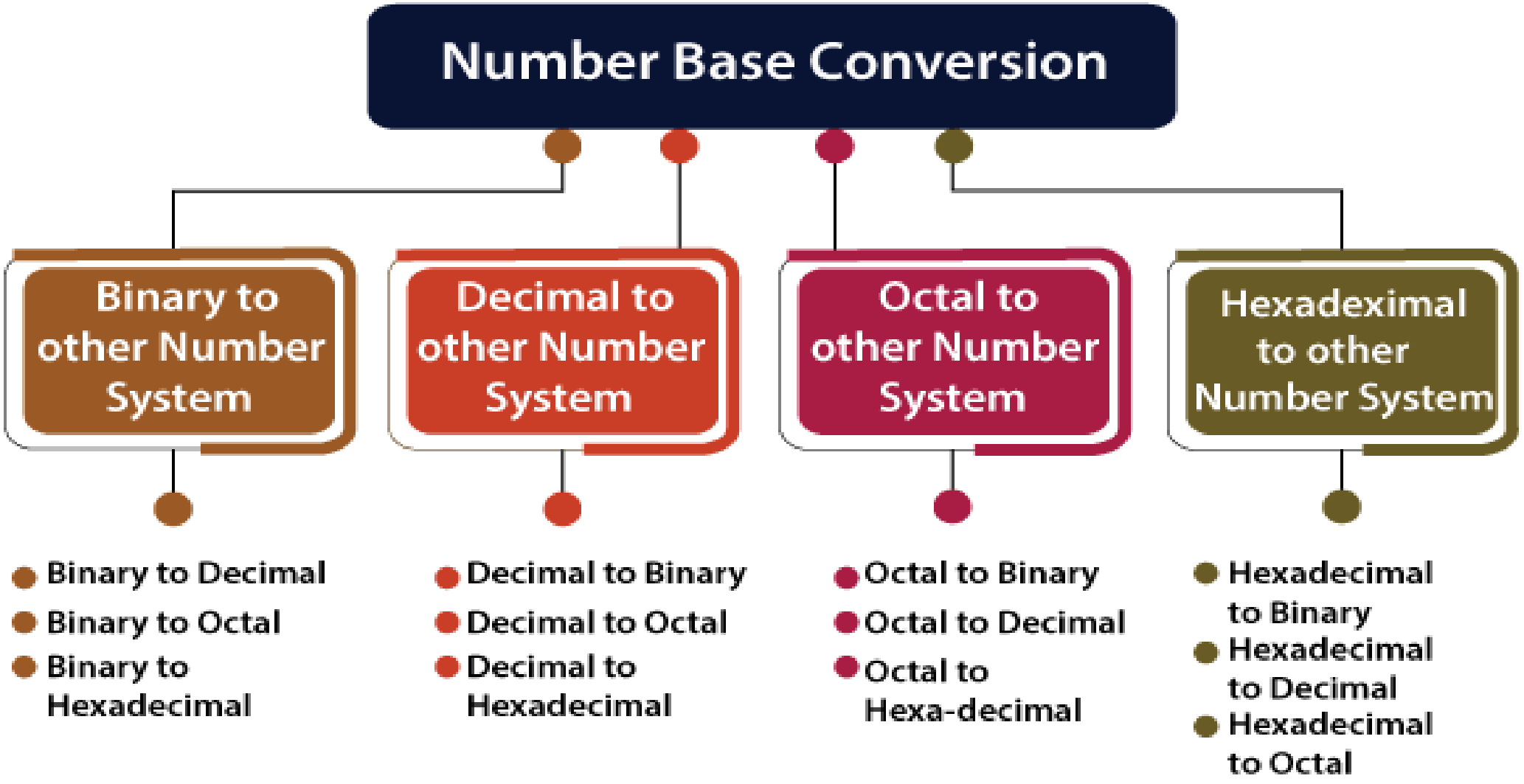
Decimal	Binary	Octal	Hexa-decimal
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Conversion Among Bases

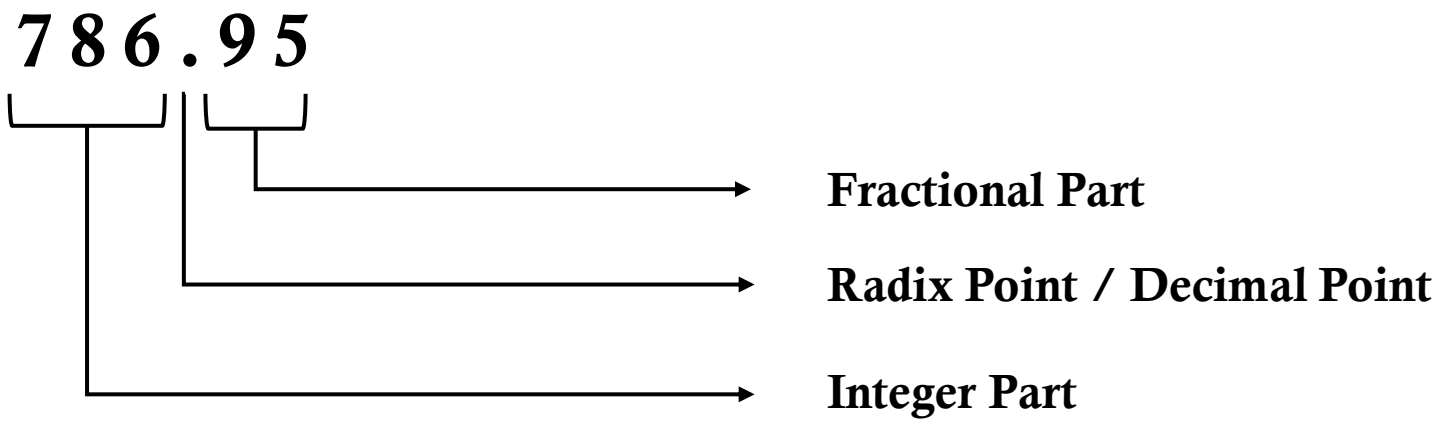
- The possibilities:



Conversion from One Number System to Another



➢ Decimal Number System



For integer part,

Digit position will be incremented from right to left

$$786 = 7 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$$

MSD (most significant digit)

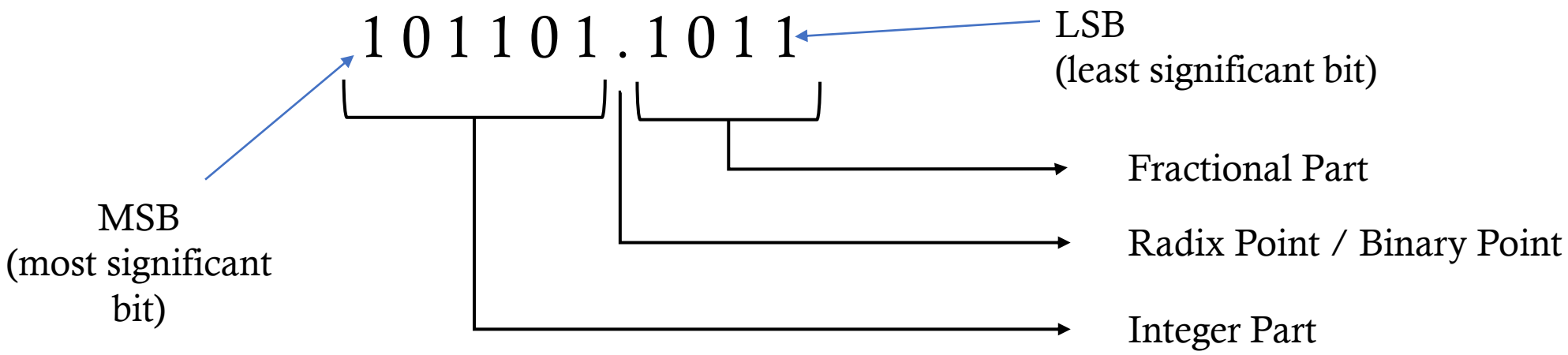
LSD (least significant digit)

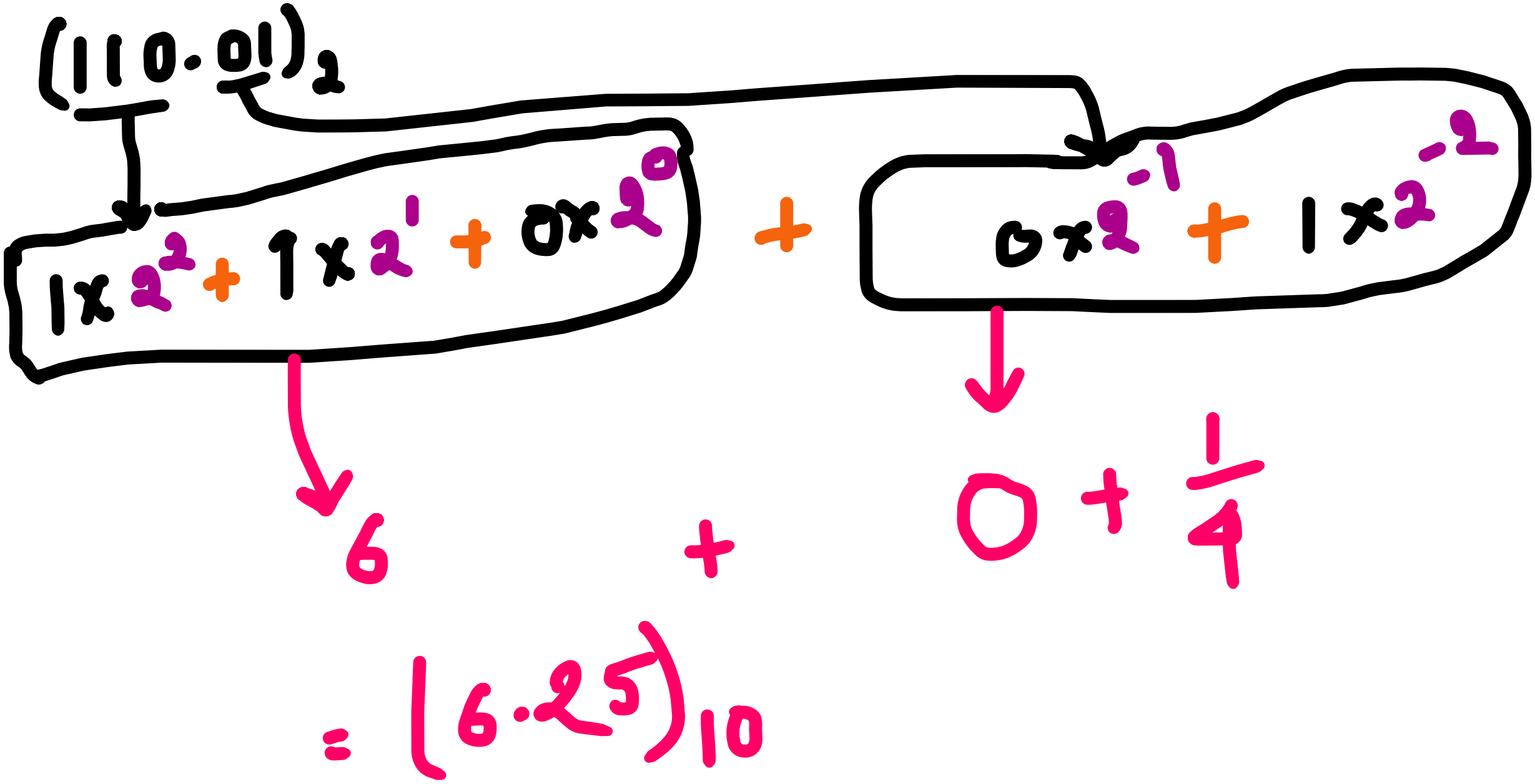
For Fractional Part,

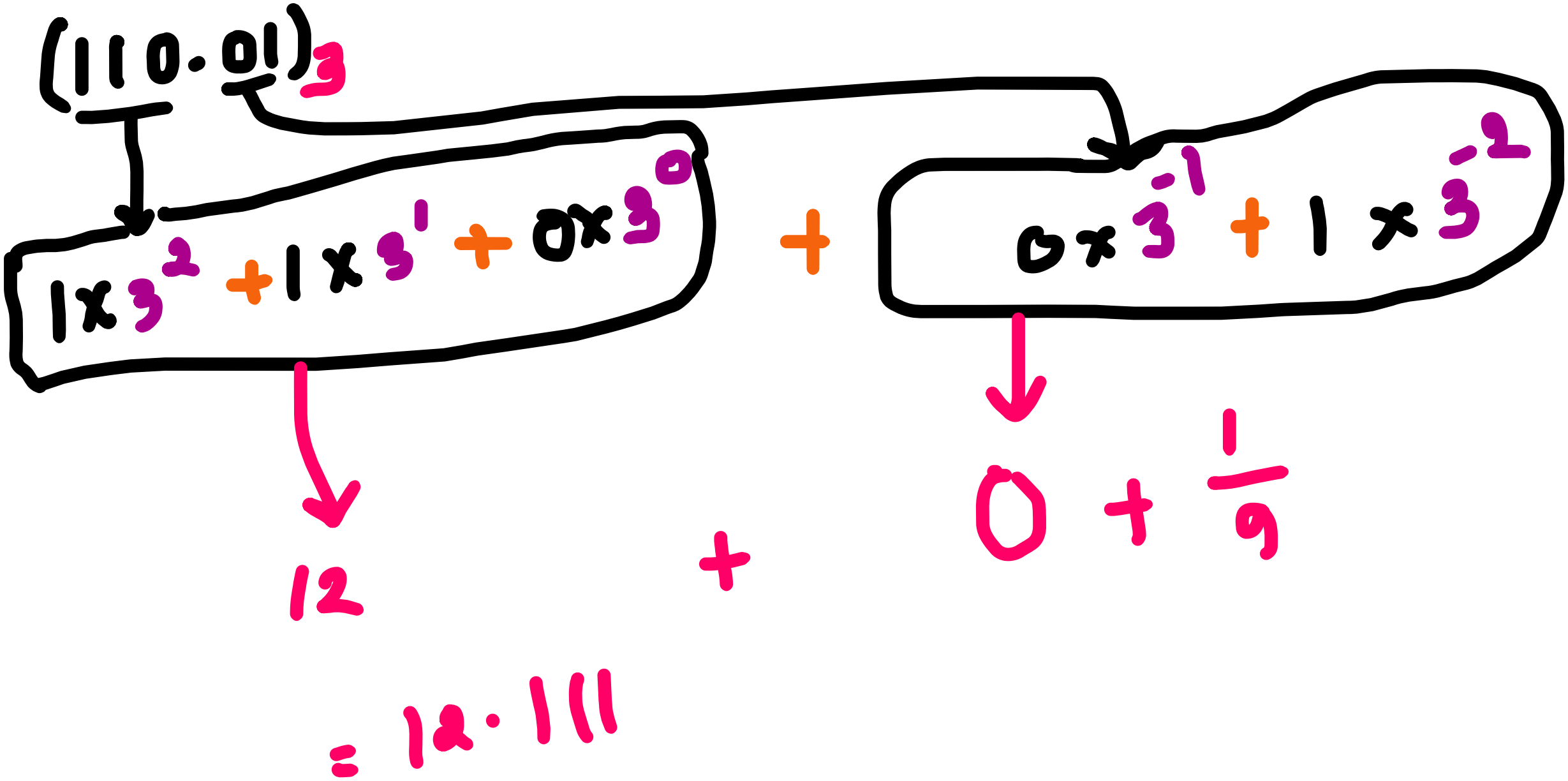
$$.95 = 9 \times 10^{-1} + 5 \times 10^{-2}$$

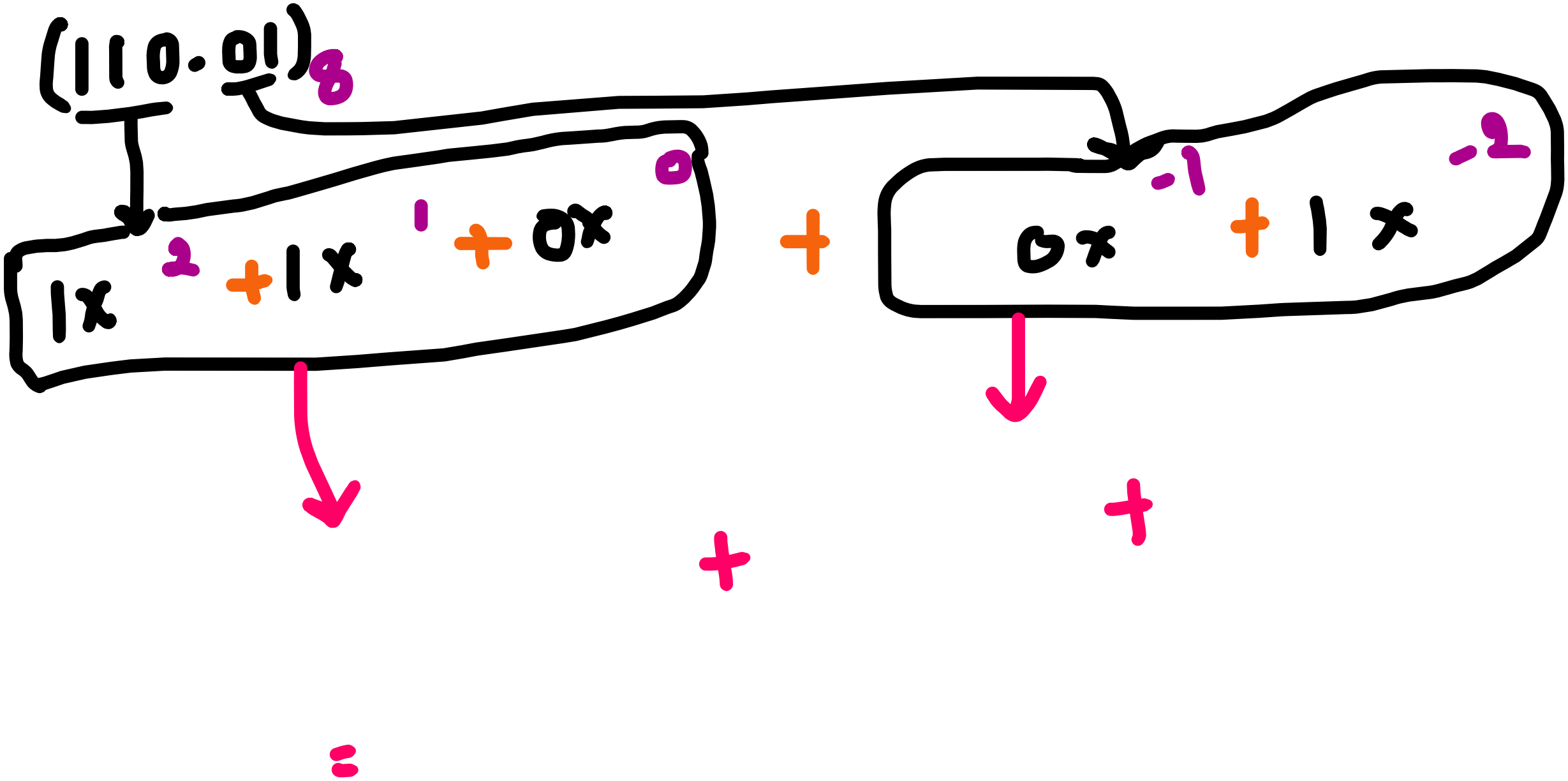
Power is decreasing from left to right

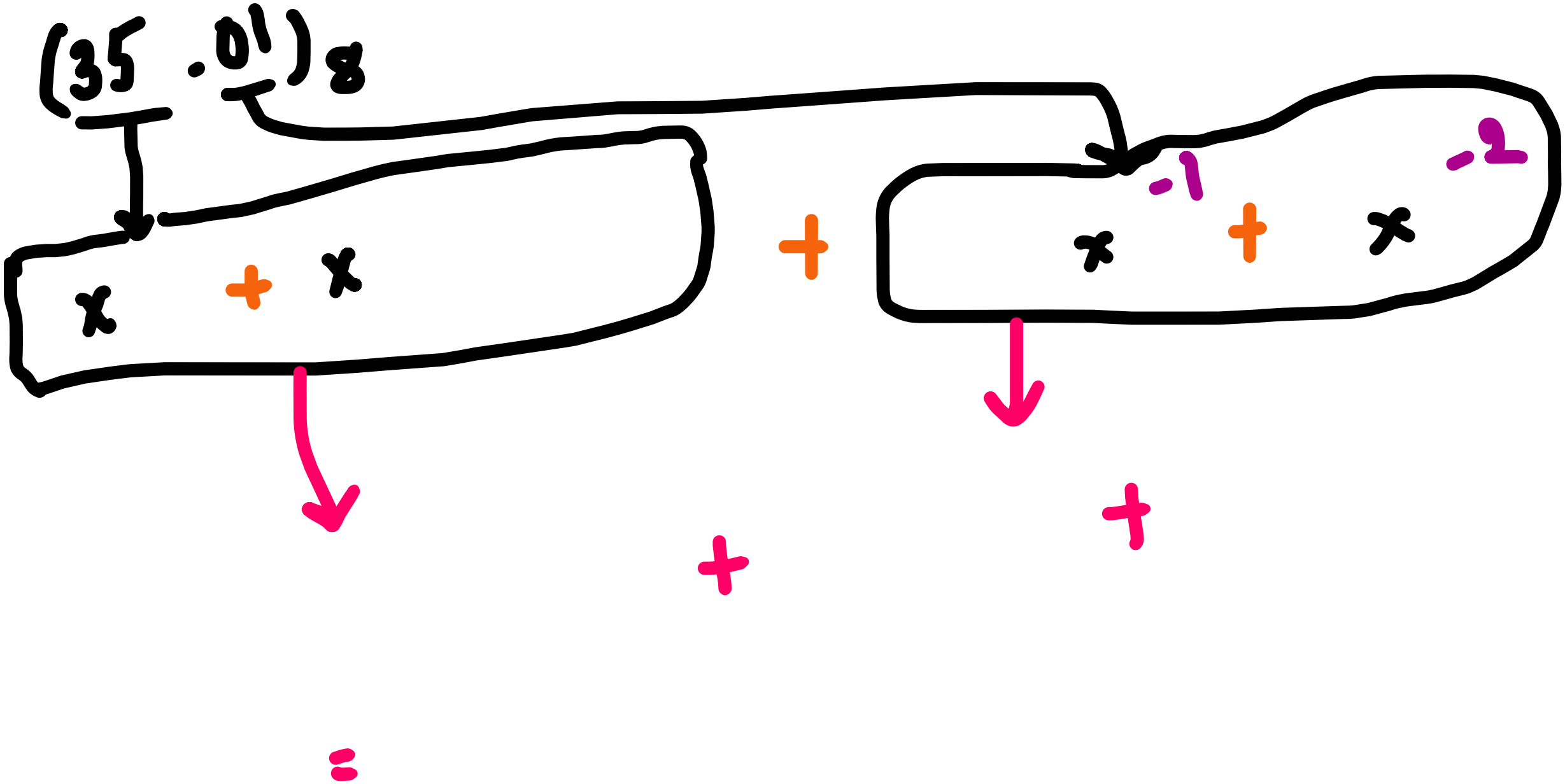
➢ Binary Number System

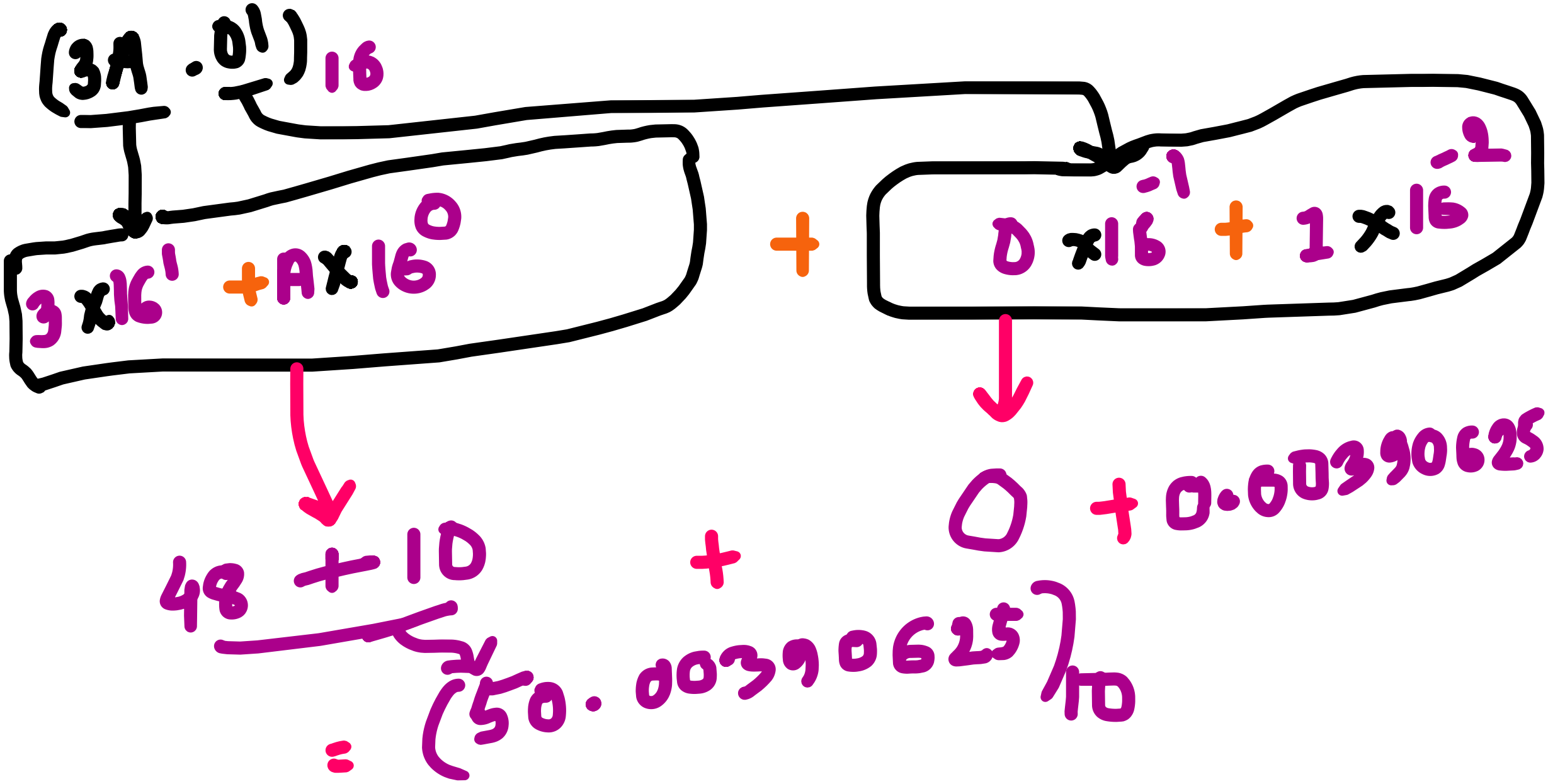



$(110.01)_2$
$1\times2^2+1\times2^1+0\times2^0$
+
$0\times2^{-1}+1\times2^{-2}$
6
+
$0+\frac{1}{4}$
$=(6.25)_{10}$


(110.01)3
1x3^2 + 1x3^1 + 0x3^0
+
0x3^-1 + 1x3^-2
12
+
0 + 1/9
= 12.111


(110.01)8
1x 2 +1x 1 +0x 0
+
0x -1 +1x -2
=


(35 .01)8
x + x
+
x -1 + x -2
+
+
=


(3A .01)16
3 x 16^1 + A x 16^0
+
0 x 16^-1 + 1 x 16^-2
48 + 10
0 + 0.00390625
= (50.00390625)10

➢ Conversion from Decimal to Non-decimal System

- For the **integer part of the decimal number**

Step 1: Divide the integer part of the decimal number by the base b of the new system (binary, octal, hexadecimal)

Step 2: Record the remainder as the LSD of the new system

Step 3: Divide the quotient again by the base b, the remainder is the next digit of the new system.

Step 4: Continue this process until the quotient is 0.

- **For fractional part:**

Step 1: Multiply the fractional part of the decimal number by the base **b** of the new system (binary, octal, hexadecimal). The product has two parts: an integer part and a fractional part.

Step 2: Record the integer part as the LSD of the fractional part of the new system

Step 3: Multiply **the fractional part** of the product again by the base b. The product has also two parts: an integer part and a fractional part.

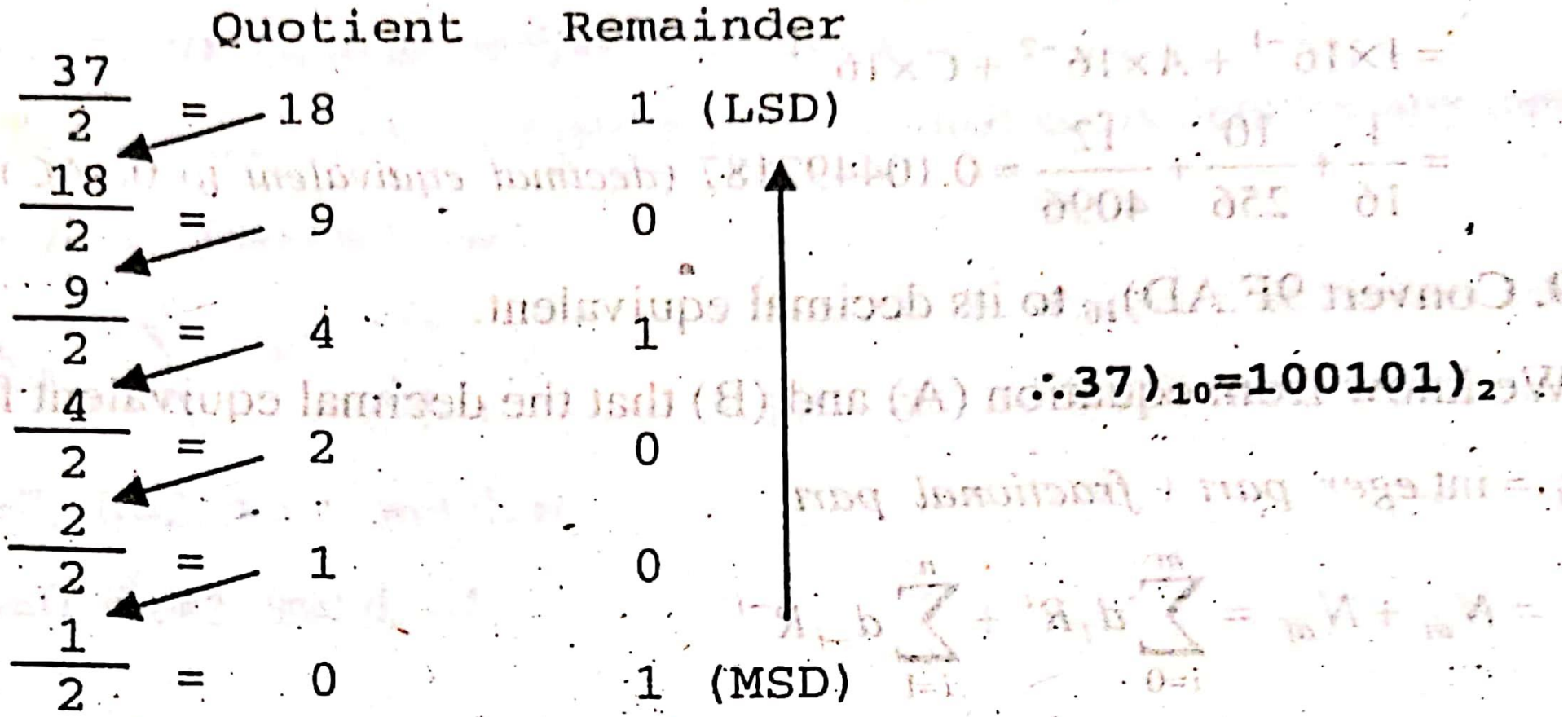
Step 4: Record this process until the **fractional part** of **a product is 0** or **a duplicate** fractional part occurs.

Step 5: Continue this process until the fractional part of a product is 0 or a duplicate fractional part occurs.

Example-03

Convert the decimal number 37 to i) a binary number, ii) an octal number, and iii) a hexadecimal number

i) **Following the above steps (here b=2 for binary number system):**



ii) $37)_{10} = ?)_8$

Following the above steps (here b=8 for octal number system):

	Quotient	Remainder
$\frac{37}{8}$ =	4	5 (LSD)
$\frac{4}{8}$ =	0	4 (MSD)

$\therefore 37)_{10} = 45)_8$

iii) $37)_{10} = ?)_{16}$

Following the above steps (here b=16 for hexadecimal number system):

	Quotient	Remainder
$\frac{37}{16}$ =	2	5 (LSD)
$\frac{2}{16}$ =	0	2 (MSD)

$\therefore 37)_{10} = 25)_{16}$

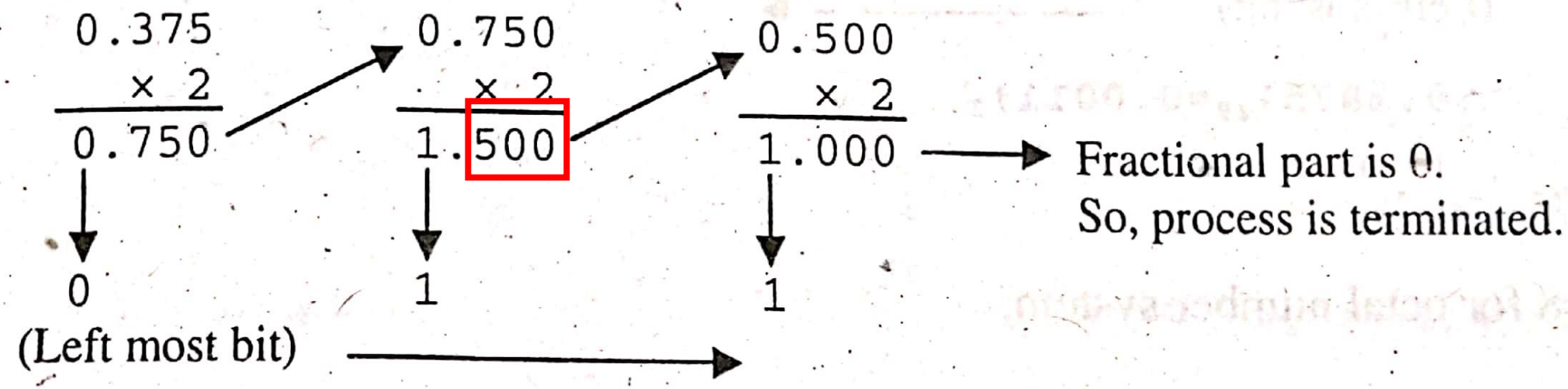
Example -04

Convert the decimal number 0.375 to i) a binary number, ii) an octal number, and iii) a hexadecimal number

Solution: i) $0.375)_{10} = ?)_2$

Following the above steps for fractional decimal part:

Here b=2 for binary number system.

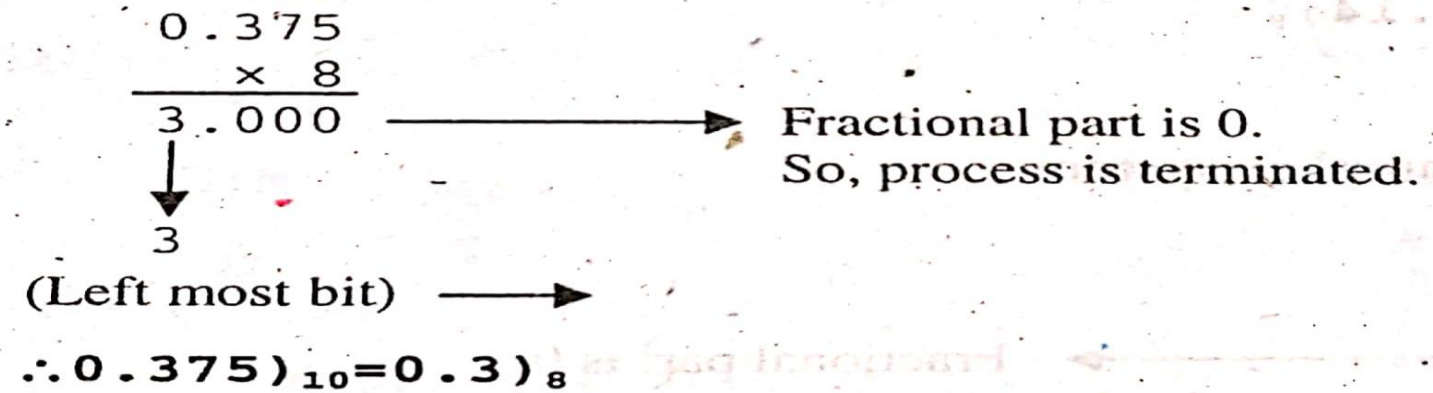


$\therefore 0.375)_{10} = 0.011)_2$

ii) $0.375)_{10} = ?)_8$

Following the above steps for fractional decimal part:

Here b=8 for octal number system.

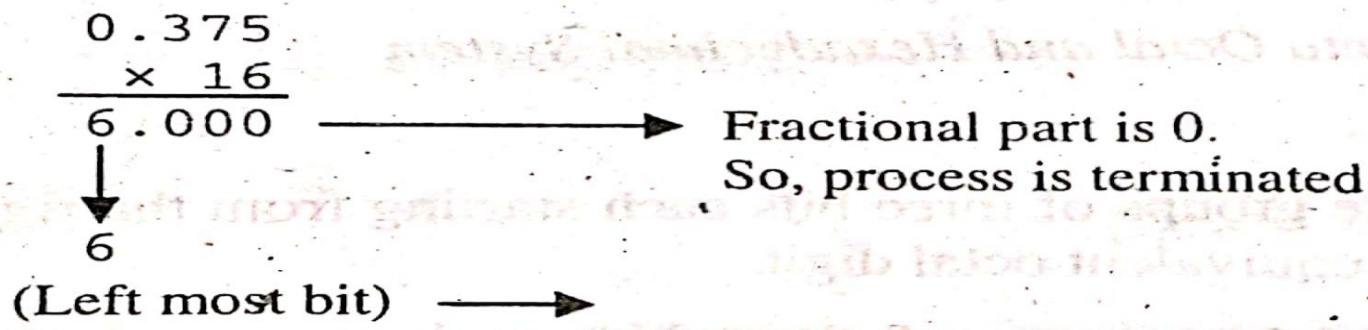


$\therefore 0.375)_{10} = 0.3)_8$

iii) $0.375)_{10} = ?)_{16}$

Following the above steps for fractional decimal part:

Here b=16 for hexadecimal number system.



$\therefore 0.375)_{10} = 0.6)_{16}$

➢ Conversion from Binary to Octal and Hexadecimal System

- **Binary-to-octal:**

a) **Integer Part:** Make groups of **three** bits each starting from the right. Then replace each group by an equivalent octal digit.

b) **Fractional Part:** Make groups of **three** bits each starting from the left. Then replace each group by an equivalent octal digit.

- **Binary-to-hexadecimal:**

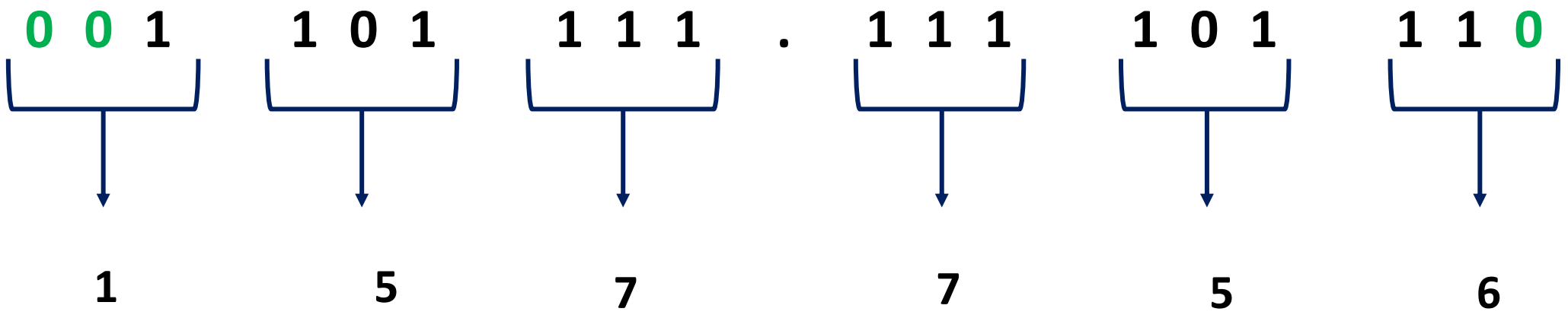
a) **Integer Part:** Make groups of **four** bits each starting from the right. Then replace each group by an equivalent hexadecimal digit.

b) **Fractional Part:** Make groups of **four** bits each starting from the left. Then replace each group by an equivalent hexadecimal digit.

Example -05

Convert the following binary number to their octal equivalent

i) 1101111.11110111



$$1101111.11110111)_2 = 157.756)_8$$

- Convert the following to the hexa-decimal numbers:
(ii) 1111101101_2

Solution:

$00\underline{11}\overline{1110}\underline{1101}$

$= 0011 \ \ 1110 \ \ 1101$

$= 3ED_{16}$

Therefore, $11\ 1110\ 1101_2 = 3ED_{16}$

- Convert the following to binary equivalents:
  (a) $A748_{16}$

Solution:

$A748_{16}$

$= 1010\ 0111\ 0100\ 1000$

$= 1010011101001000_2$

Convert the following to binary equivalents:
(b) $BA2.23C_{16}$

- **Solution:**

  $BA2.23C_{16}$

  $= 1011\ 1010\ 0010\ .\ 0010\ 0011\ 1100_2$

  $= 101110100010.00100011111_2$

➢ Conversion from Octal to Hexadecimal System and Vice Versa

- **Octal to hexadecimal:**

a) Convert the octal number to its equivalent binary number

b) Then convert the binary number to hexadecimal number

- **Hexadecimal to Octal:**

a) Convert the hexadecimal number to its equivalent binary number

b) Then convert the binary number to octal number

➢ Convert 1573_8 to hexa-decimal

- **Solution:**

1573_8

$= 001\ 101\ 111\ 011$

$= 0011\ 0111\ 1011$

$= 37B_{16}$

Hence $1573_8 = 37B_{16}$

Exercise….

Decimal	Binary	Octal	Hexa-decimal
33			
	1110101		
		703	
			1AF

Don’t use a calculator!

Exercise – Convert …

Answer

Decimal	Binary	Octal	Hexa-decimal
33	100001	41	21
117	1110101	165	75
451	111000011	703	1C3
431	110101111	657	1AF

Exercise

Decimal	Binary	Octal	Hexa-decimal
29.8			
	101.1101		
		3.07	
			C.82

Exercise – Convert …

Answer

Decimal	Binary	Octal	Hexa-decimal
29.8	11101.110011…	35.63…	1D.CC…
5.8125	101.1101	5.64	5.D
3.109375	11.000111	3.07	3.1C
12.5078125	1100.10000010	14.404	C.82

Bits and Bytes

- All of the data stored and transmitted by digital devices is encoded as **bits**.
- Terminology related to bits and bytes is extensively used to describe storage capacity and network access speed.
- The word **bit**, an abbreviation for binary digit, can be further abbreviated as a lowercase b.
- A **group of eight bits** is called a **byte** and is usually abbreviated as an uppercase B.
- When reading about digital devices, you'll frequently encounter references such as 90 kilobits per second, 1.44 megabytes, 2.8 gigahertz, and 2 terabytes.
- Kilo, mega, giga, tera, and similar terms are used to quantify digital data.

	Approx. Bytes	**Actual Bytes**	**Approx. Bits**	**Typical file/media**
1B	1	1	8	Text email, SMS
1KB	1000 B = 10^3	1024 B = 2^{10}	8 x 10^3	Word document
1MB	1000 KB = 10^6	1024 KB = 2^{20}	8 x 10^6	Digital Photo
1GB	1000 MB = 10^9	1024 MB = 2^{30}	8 x 10^9	DVD
1TB	1000 GB = 10^{12}	1024 GB = 2^{40}	8 x 10^{12}	Hard disk
1PB	1000 TB = 10^{15}	1024 TB = 2^{50}	8 x 10^{15}	Cloud!

Memory Capacity Conversion Chart

Term (Abbreviation)	Approximate Size
Byte (B)	8 bits
Kilobyte (KB)	1024 bytes / 10^3 bytes
Megabyte (MB)	1024 KB / 10^6 bytes
Gigabyte (GB)	1024 MB / 10^9 bytes
Terabyte (TB)	1024 GB / 10^{12} bytes
Petabyte (PB)	1024 TB / 10^{15} bytes
Exabyte (EB)	1024 PB / 10^{18} bytes
Zettabyte (ZB)	1024 EB / 10^{21} bytes
Yottabyte (YB)	1024 ZB / 10^{24} bytes

- Use bits for data rates, such as Internet connection speeds, and movie download speeds.
- Use bytes for file sizes and storage capacities.
- **104 KB**: Kilobyte (KB or Kbyte) is often used when referring to the size of small computer files.
- **56 Kbps**: Kilobit (Kb or Kbit) can be used for slow data rates, such as a 56 Kbps (kilobits per second) dial-up connection.
- **50 Mbps**: Megabit (Mb or Mbit) is used for faster data rates, such as a 50 Mbps (megabits per second) Internet connection.
- **3.2 MB**: Megabyte (MB or MByte) is typically used when referring to the size of files containing photos and videos.
- **100 Gbit**: Gigabit (Gb or Gbit) is used for really fast network speeds.
- **16 GB**: Gigabyte (GB or GByte) is commonly used to refer to storage capacity.